A NON-JACOBSON KOLMOGOROV SPACE WITH A BASIS OF QUASI-COMPACT OPENS HAS A LOCALLY CLOSED NON-CLOSED POINT

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Allowed standard prerequisites (editorial).

- The statement that a quasi-compact space has a closed point whose complement is controlled by the standing hypotheses. Invoked inside the counted proof (source lines 3583-3585). Produces a point closed in the relevant closed subset.

Source reference. Original lemma lemma-quasi-compact-closed-point: sources/stacks-topology/pinned-original-source.tex, source lines 1985–1985. Navigation pointer only: the counted text refers to the Stacks item lemma-quasi-compact-closed-point, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let X be a Kolmogorov topological space with a basis of quasi-compact open sets. If X is not Jacobson, then there exists a non-closed point $x \in X$ such that $\{x\}$ is locally closed.*

Proof. As X is not Jacobson there exists a closed set Z and an open set U in X such that $Z \cap U$ is nonempty and does not contain points closed in X. As X has a basis of quasi-compact open sets we may replace U by an open quasi-compact neighborhood of a point in $Z \cap U$ and so we may assume that U is quasi-compact open. By Lemma lemma-quasi-compact-closed-point, there exists a point $x \in Z \cap U$ closed in $Z \cap U$, and so $\{x\}$ is locally closed but not closed in X. □

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